

Community detection in geometric graphs: Recent advances until 2026

Raphaël Sala

Affiliation à renseigner
`raphael.sala@utoulouse.fr`

Résumé TODO

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1 Introduction

Community detection consists in finding clusters in a graph where nodes share more connections with nodes of the same cluster than with nodes of other clusters. This broad topic is detailed in [Fortunato, 2010]. In this report, we focus on random geometric graphs (RGG). Adding spatial information allows us to consider local information based on real or abstract proximity. In social networks, two individuals are more likely to be friends if they are geographically close, or if they share similar interests, age, ... More concrete examples of RGG can be found in wireless networks [Gomez et al., 2015], in biology [Higham et al., 2008], and also in community detection problems [Sankararaman et al., 2020] word embedding, physic,

The main reference model for community detection is the Stochastic Block Model (SBM) [Holland et al., 1983]. Each node carries a hidden community label and edge probabilities depend on whether two nodes belong to the same community or not. This model comes from the Erdős-Rényi graph, where each node connects to others with the same probability [Erdős et al., 1960]. This model is now well understood, see [Abbe, 2018]. Nonetheless, it has some drawbacks, since it is unrealistic to think that every node can connect to every other one with the same probability scale. Around the same time as Erdős-Rényi, the RGG was introduced by [Gilbert, 1961]. Add geometric information allows us to mimic some emergent behaviours of real networks, such as sparsity [Del Genio et al., 2011], the "small-world" property [Watts and Strogatz, 1998] or transitivity, namely the idea that a friend of a friend is more likely to be a friend.

RGG and its properties are well studied, see [Penrose, 2003]. When we add community structure, we are in a different setting. We observe the graph, and sometimes also the node locations, while the community labels remain hidden. The natural question is then whether the latent structure can be detected, and if so, how accurately the labels can be recovered. We are interested in several levels

of inference. First, *distinguishability* asks whether the observed graph comes from a geometric model with communities or from a null model without community structure. This is a testing problem. It aims to distinguish the presence of a community structure, not to recover the labels. Then, if recovery is possible, one can ask how accurate it can be.

- *Weak/partial recovery* : detecting communities better than random guessing
- *Almost exact recovery* : the fraction of misclassified vertices vanishes as the number of nodes grows.
- *Exact recovery* : all community labels are correctly recovered, up to permutation.

Thus, in this report, community detection is not only understood as producing a convenient partition of the observed graph, but rather as recovering latent community labels under a probabilistic geometric model.

Previously, we emphasized properties that RGG share with real networks due to the geometric information. However, one can also obtain transitivity, and surely more, with attribute-based models [Li et al., 2018], hypergraphs [Ruggeri et al., 2023] and even geometric hypergraphs [Kovács et al., 2025]. We will not consider these models here, in order to keep the focus on the geometric aspect. Moreover, among the problems mentioned above, one is *Distinguishability* which should not be confused with the *Detection Problem* that we do not study here. The latter usually asks for conditions on the parameters defining the RGG under which the model is distinguishable from an Erdős-Rényi graph. Hence, this is not directly a community question. See, [Bubeck et al., 2016] [Liu et al., 2022] [Mao et al., 2026]. Finally, at the beginning of the literature, some works were interested in recovering the latent position of nodes in the geometric space, and considered this as a community detection problem, [Valdivia and De Castro, 2019] [Eldan et al., 2022]. This is a different problem from the one we are interested in, and it's in some sense harder, because one wants to recover the position (a continuous label) of each node rather than a community (a discrete label).

2 Modèle

- pas de boucle, pas dirige, pas de communautés chevauchantes, etc...
- espace latent (ou pas) [Castro et al., 2020]
- sphère, espace euclidien ?
- régime : dense, logarithmique, sparse
- hard-RGG, soft-RGG [Duchemin and De Castro, 2023]
- motivation : bien que les modèles ne soient pas les mêmes, ils n'ont pas pour but de comprendre tous ceux possible mais qu'à travers chaque classe voir les mécanismes. Exemple : passer du local au global puis raffinement, les quantités utilisées, ...

Le but est de faire des notations claires, concises et générales pour pouvoir expliquer tous les modèles sans avoir à se perdre dans la notation.

3 Seuil théorique information

Faire le rapport dans l'ordre chronologique je pense

- [Galhotra et al., 2018] (Geometric Block Model, GBM)
- [Sankararaman and Baccelli, 2018] & [Abbe et al., 2021] (Geometric SBM)
- [Chien et al., 2020]
- [Galhotra et al., 2020] (GBM)
- [Galhotra et al., 2023] (GBM)
- [Avrachenkov et al., 2024] (Geometric Kernel Block Model, GKBM)
- [Gaudio et al., 2024b] (Geometric SBM)
- [Gaudio et al., 2024a] (Geometric Hidden Community Model, GHCM)
- [Gaudio and Guan, 2025] (GHCM)
- [Gaudio and Jin, 2026] (GHCM)

3.1 Geometric Block Model (GBM)

[Galhotra et al., 2018] [Galhotra et al., 2020] [Galhotra et al., 2023]
[Chien et al., 2020]

3.2 Geometric Stochastic Block Model (GSBM)

[Sankararaman and Baccelli, 2018] [Abbe et al., 2021]

3.3 (Geometric Kernel Block Model, GKBM)

[Avrachenkov et al., 2024] [Allem et al., 2025]

3.4 Geometric Hidden Community Model (GHCM)

[Gaudio et al., 2024b] [Gaudio et al., 2024a] [Gaudio and Guan, 2025] [Gaudio and Jin, 2026]

4 Spectral

(A confirmer) [Chen et al., 2016]
(papier SBM mais théorie intéressante) [Bordenave et al., 2015]
Cadre spectral intéressant pour certains espaces [Méliot, 2019]
Méthode spectrale + Robustesse (la perturbation est géométrique) [Péché and Perchet, 2020]
Méthode spectrale sur Soft Geometric Block Model (SGBM) (Generalisation de Geometric SBM) [Avrachenkov et al., 2021]

5 Ouvertures

- Considérer des modèles hyperboliques, pas seulement labels & distance-dependant, car ceux-ci captent mieux les propriétés habituelles des réseaux
- Clustering [Krioukov, 2016]
- Degre inhomogène [Barabási and Albert, 1999]
- "Communities with specific mixing patterns" (mesoscale) [Toivonen et al., 2006]

Communautés apparaissent dans un espace hyperbolique latent [Bruno et al., 2019] Réponse 6ans plus tard : pas toujours! + definition d'un modele pour "régler" ça [Guarino et al., 2025]
[Li et al., 2018]

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